

TABLE I. DATASET AFTER AUGMENTATION BEACON 1 AND 2

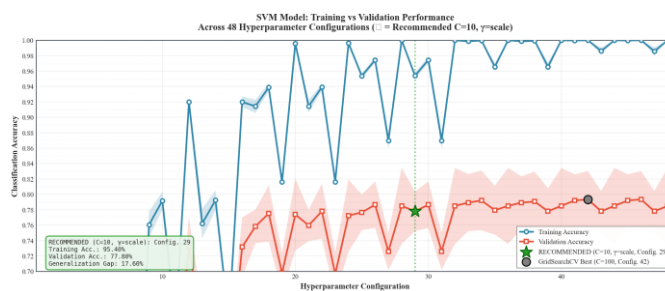
Area	B1_median	B1_std	B1_iqr	B2_median	B2_std	B2_iqr
Area_G	-77.04	5.75	8.10	-67.88	2.26	3.16
Area_D	-75.36	5.68	8.20	-72.48	5.43	6.95
Area_I	-67.88	7.01	10.43	-75.35	4.51	8.97
Area_H	-72.41	3.54	5.78	-69.99	3.08	6.40
Area_E	-68.00	4.79	6.00	-70.00	5.32	7.00
Area_C	-54.04	2.71	1.80	-77.75	5.16	6.72
Area_C	-100.00	0.00	0.00	-77.42	4.21	3.71
Area_C	-59.00	5.33	10.00	-71.00	2.82	3.00
Area_B	-71.11	6.48	5.50	-64.85	2.50	4.42
Area_A	-74.36	3.54	7.22	-52.96	4.85	5.23

TABLE II. DATASET AFTER AUGMENTATION BEACON 3 AND 4

Area	B3_median	B3_std	B3_iqr	B4_median	B4_std	B4_iqr
Area_G	-53.78	1.83	2.88	-73.36	2.85	2.94
Area_D	-66.71	3.63	4.29	-79.66	5.43	8.32
Area_I	-100.00	0.00	0.00	-53.05	2.91	3.73
Area_H	-100.00	0.00	0.00	-71.65	7.15	6.73
Area_E	-71.00	5.26	7.00	-65.00	6.52	14.00
Area_C	-77.20	4.32	4.95	-100.00	0.00	0.00
Area_C	-100.00	0.00	0.00	-70.07	4.71	6.25
Area_C	-100.00	0.00	0.00	-71.00	5.64	11.00
Area_B	-78.90	3.04	4.07	-69.23	3.65	3.16
Area_A	-70.99	1.94	2.20	-71.02	3.32	3.03

TABLE III. DATASET AFTER AUGMENTATION BEACON 5

Area	B5_median	B5_std	B5_iqr
Area_G	-81.13	3.44	3.90
Area_D	-100.00	0.00	0.00
Area_I	-66.31	6.45	11.71
Area_H	-55.18	4.70	2.27
Area_E	-52.00	2.55	3.00
Area_C	-100.00	0.00	0.00
Area_C	-73.06	3.39	3.02
Area_C	-69.00	1.53	2.00
Area_B	-100.00	0.00	0.00
Area_A	-71.00	4.63	3.89



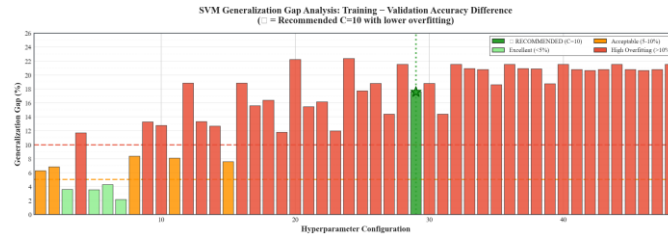


Fig. 1. Comparison of SVM training and validation testing

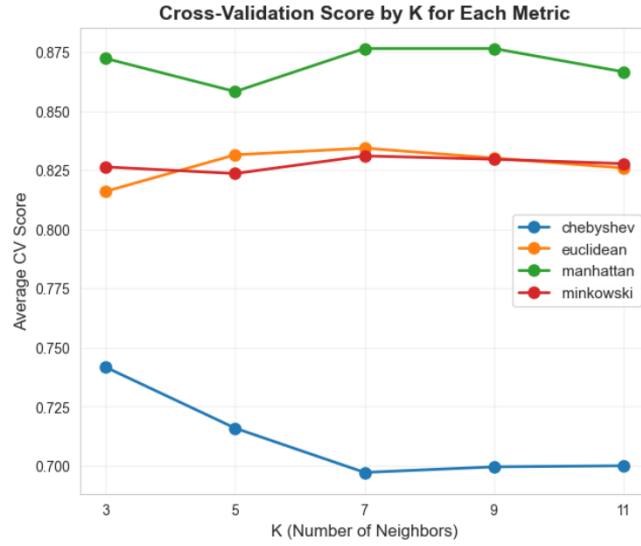


Fig. 2. Validation test with different parameter values

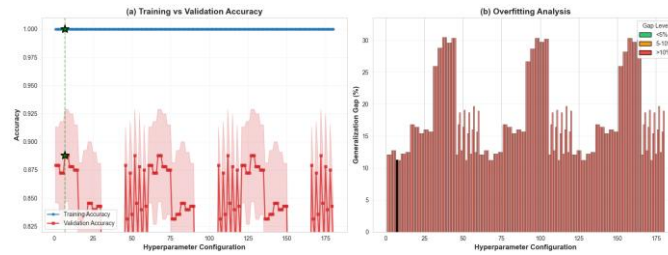


Fig. 3. Hyperparameter training and validation results graph and overfitting

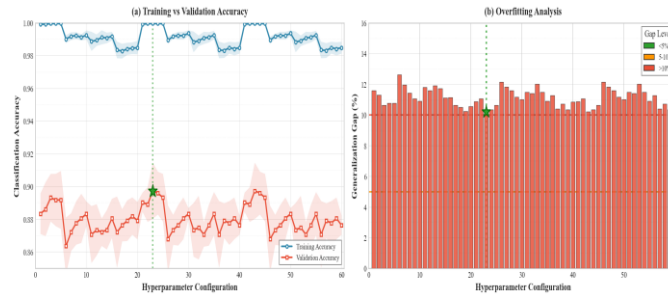


Fig. 4. Comparison of hyperparameter configuration and overfitting gap

TABLE IV. REAL-TIME TRACKING RESULTS WITH ALL BEACONS ACTIVE

No	Actual Area	Predicted Area	Confidence	Time (ms)	Prediction Result
1	Area_A	A	65.42%	168.92	TRUE
2	Area_A	A	57.90%	168.9	TRUE
3	Area_A	A	78%	168.9	TRUE
4	Area_B	B	20.12%	169.01	TRUE
5	Area_B	B	62.91%	169.05	TRUE
6	Area_B	B	29.63%	169.06	TRUE
7	Area_C	C	79.62%	169.07	TRUE
8	Area_C	C	83.84%	169	TRUE
9	Area_C	C	80.40%	169	TRUE
10	Area_D	D	43.28%	169	TRUE
11	Area_D	B	22.25%	169	FALSE
12	Area_D	D	37.06%	169	TRUE
13	Area_E	E	68.45%	169	TRUE
14	Area_E	E	86.59%	169	TRUE
15	Area_E	E	93.64%	169	TRUE
16	Area_F	F	47.96%	149	TRUE
17	Area_F	F	37.08%	140.53	TRUE
18	Area_F	F	36.49%	133.77	TRUE
19	Area_G	G	92.30%	117.29	TRUE
20	Area_G	G	93.32%	118.23	TRUE
21	Area_G	G	93%	119.24	TRUE
22	Area_H	H	49.05%	123.3	TRUE
23	Area_H	B	40.50%	122.86	FALSE
24	Area_H	H	56.26	122.56	TRUE
25	Area_I	I	60.47	125.93	TRUE
26	Area_I	I	70.37%	125.08	TRUE
27	Area_I	I	67.82%	124.73	TRUE

TABLE V. REAL-TIME TRACKING RESULTS WITH 3 BEACONS DISCONNECTED

No	Actual Area	Predicted Area	Confidence	Time (ms)	Prediction Result
1	Area_A	A	73.06%	126.57	TRUE
2	Area_A	A	68.55%	126.44	TRUE
3	Area_A	B	43.52%	126.16	FALSE
4	Area_B	B	59.33%	87.56	TRUE
5	Area_B	B	42.08%	86.04	TRUE
6	Area_B	B	49.01%	84.43	TRUE
7	Area_C	C	28.54%	126.98	TRUE
8	Area_C	C	19.23%	126.85	TRUE
9	Area_C	D	23.38%	126.72	FALSE
10	Area_D	A	46.41%	124.57	FALSE
11	Area_D	D	46.55%	124.48	TRUE
12	Area_D	A	53.24%	124.38	FALSE
13	Area_E	E	78.84%	123.05	TRUE
14	Area_E	E	79.42%	123.14	TRUE
15	Area_E	E	79.26%	123.2	TRUE
16	Area_F	F	28.07%	123.27	TRUE
17	Area_F	C	30.16%	123.3	FALSE
18	Area_F	B	18.21%	123.34	FALSE
19	Area_G	G	75.74%	124.12	TRUE
20	Area_G	G	97.46%	124.2	TRUE
21	Area_G	G	71.58%	124.15	TRUE
22	Area_H	H	69.68%	123.85	TRUE
23	Area_H	H	72.63%	123.9	TRUE
24	Area_H	H	84.95%	123.95	TRUE
25	Area_I	I	59.57%	124	TRUE
26	Area_I	I	57.98%	124	TRUE
27	Area_I	I	70.25%	124	TRUE

TABLE VI. REAL-TIME TRACKING RESULTS WITH 1BEACONS DISCONNECTED

No	Actual Area	Predicted Area	Confidence	Time (ms)	Prediction Result
1	Area_A	A	42.27%	124.15	TRUE
2	Area_A	A	46.07%	124.11	TRUE
3	Area_A	A	42.45%	124.06	TRUE
4	Area_B	B	34.89%	125.6	TRUE
5	Area_B	B	50.94%	125.66	TRUE
6	Area_B	A	38.35%	125.62	FALSE
7	Area_C	C	72.86%	125.67	TRUE
8	Area_C	C	44.10%	125.74	TRUE
9	Area_C	C	50.45%	125.8	TRUE
10	Area_D	D	44.46%	125.96	TRUE
11	Area_D	D	48.13%	125.94	TRUE
12	Area_D	D	52.74%	125.89	TRUE
13	Area_E	E	84.22%	125.66	TRUE
14	Area_E	E	77.87%	125.62	TRUE
15	Area_E	E	91.82%	125.67	TRUE
16	Area_F	F	40.44%	137.85	TRUE
17	Area_F	F	35.12	126.17	TRUE
18	Area_F	C	38.03%	126.18	FALSE
19	Area_G	G	20.66%	125.98	TRUE
20	Area_G	G	24.44%	125.98	TRUE
21	Area_G	H	33.96%	126	FALSE
22	Area_H	G	31.63%	125.19	FALSE
23	Area_H	H	25.31%	125.2	TRUE
24	Area_H	H	22.17%	125.3	TRUE
25	Area_I	I	65.83%	124.87	TRUE
26	Area_I	I	62.73%	124.93	TRUE
27	Area_I	I	65.48%	124.98	TRUE